TASK 4 -RESPONSIBLE AI AND MODEL INTERPRETATION USING SHAP ANALYSIS REPORT

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INTRODUCTION

Responsible Artificial Intelligence helps ensure that machine learning systems are transparent, fair, and understandable. In this project, a machine learning model was developed using the Breast Cancer dataset and Random Forest algorithm. SHAP analysis was applied to understand feature importance and explain model predictions. Bias analysis was also performed to evaluate fairness and mitigation strategies were proposed.

OBJECTIVES

• To build and train a machine learning model

• To evaluate model performance

• To analyze feature importance using SHAP

• To identify possible bias in predictions

• To provide mitigation techniques for responsible AI

TOOLS AND TECHNOLOGIES USED

• Python

• Google Colab

• Pandas

• NumPy

• Matplotlib

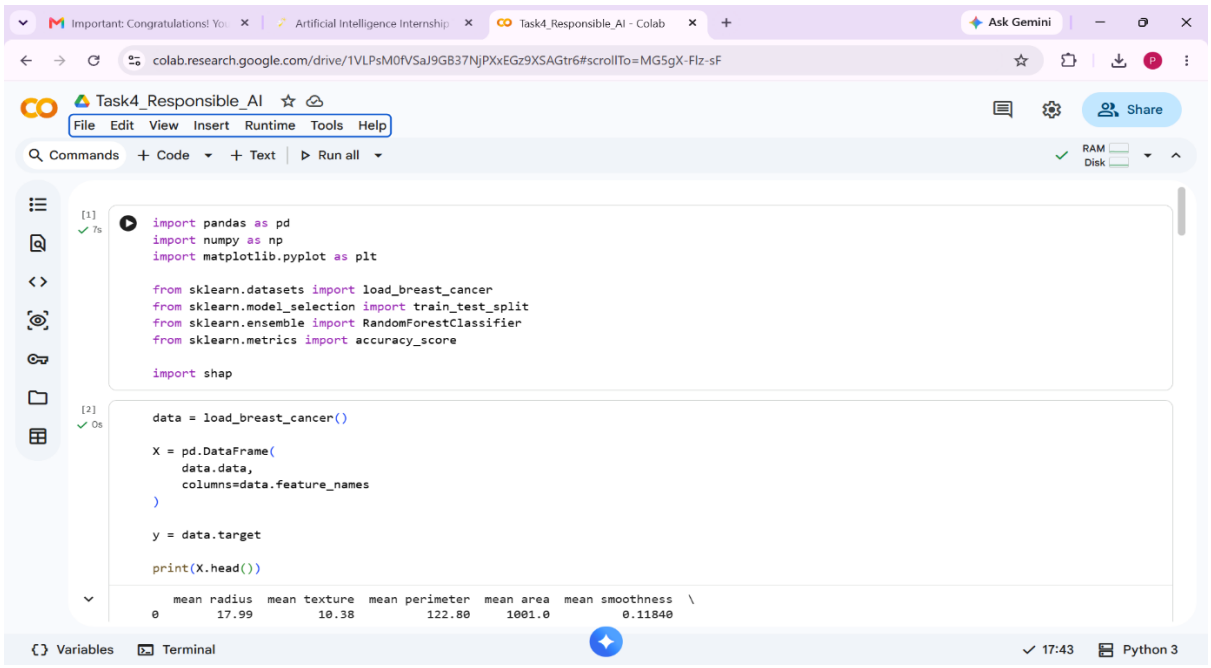
• Scikit-learn

• SHAP

IMPLEMENTATION

1. Importing Required Libraries

The necessary Python libraries were imported to perform machine learning tasks. Pandas and NumPy were used for data handling, Matplotlib for visualization, Scikit-learn for model building, and SHAP for model interpretation.



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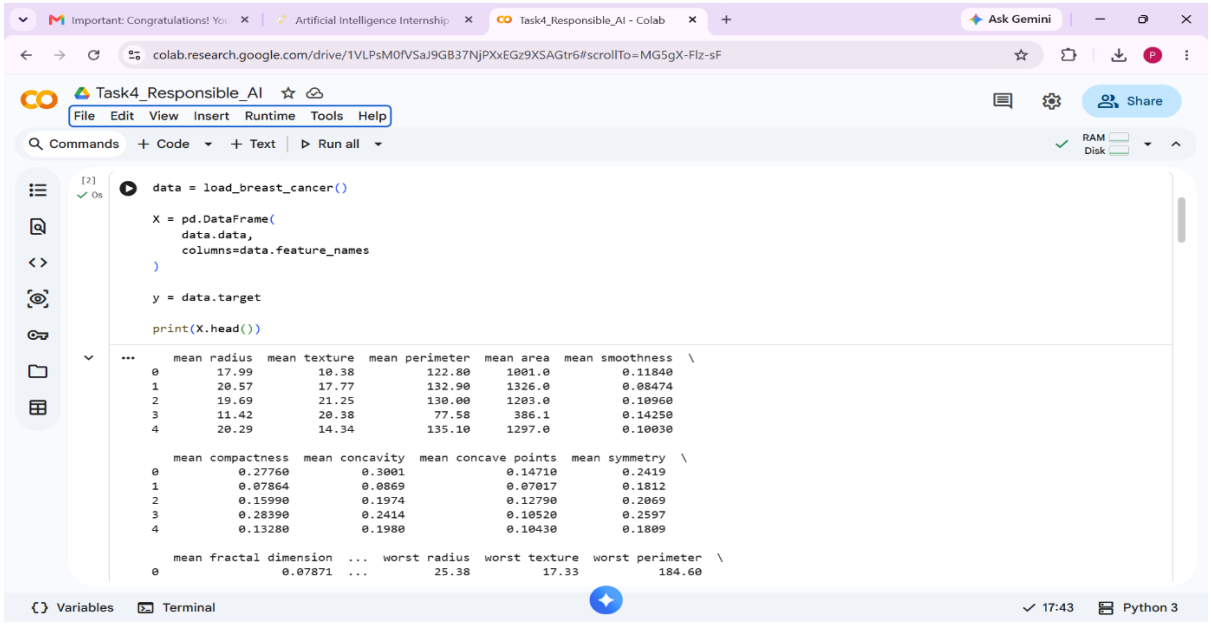
2. Loading the Dataset

The Breast Cancer dataset was loaded into the notebook. The dataset contains several medical measurements that are used to classify whether a tumor is benign or malignant.



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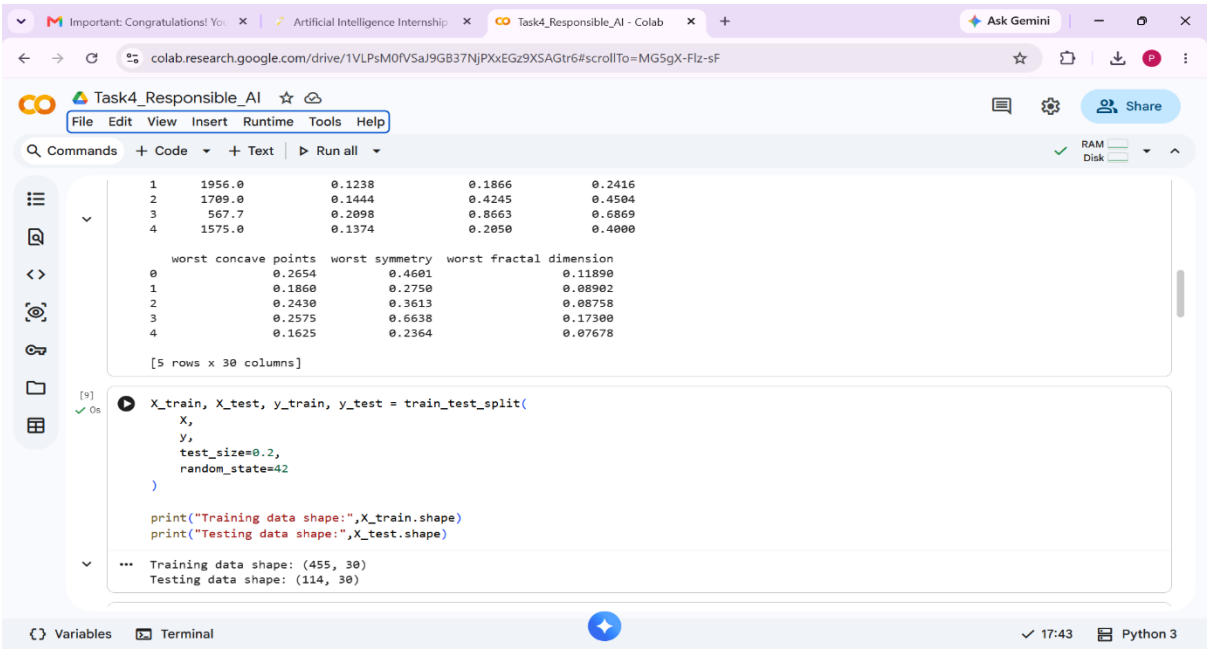
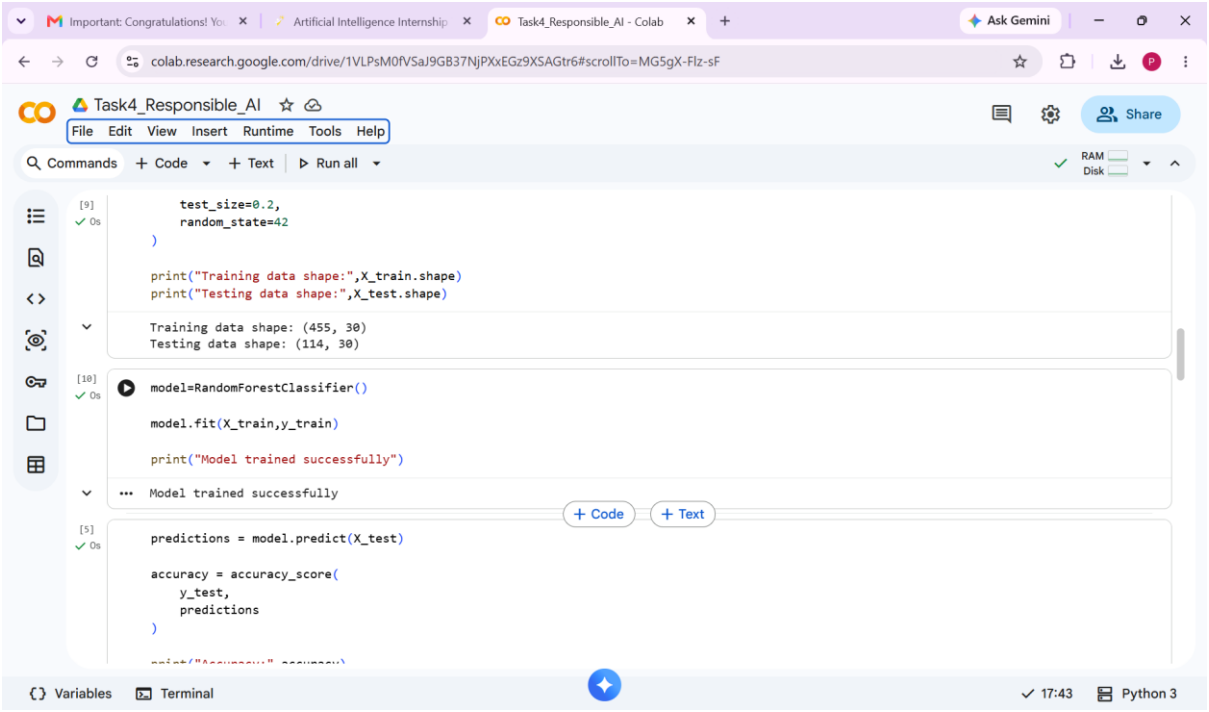
3. Splitting the Dataset

The dataset was divided into training and testing data. Training data was used to train the model while testing data was used to evaluate prediction performance.



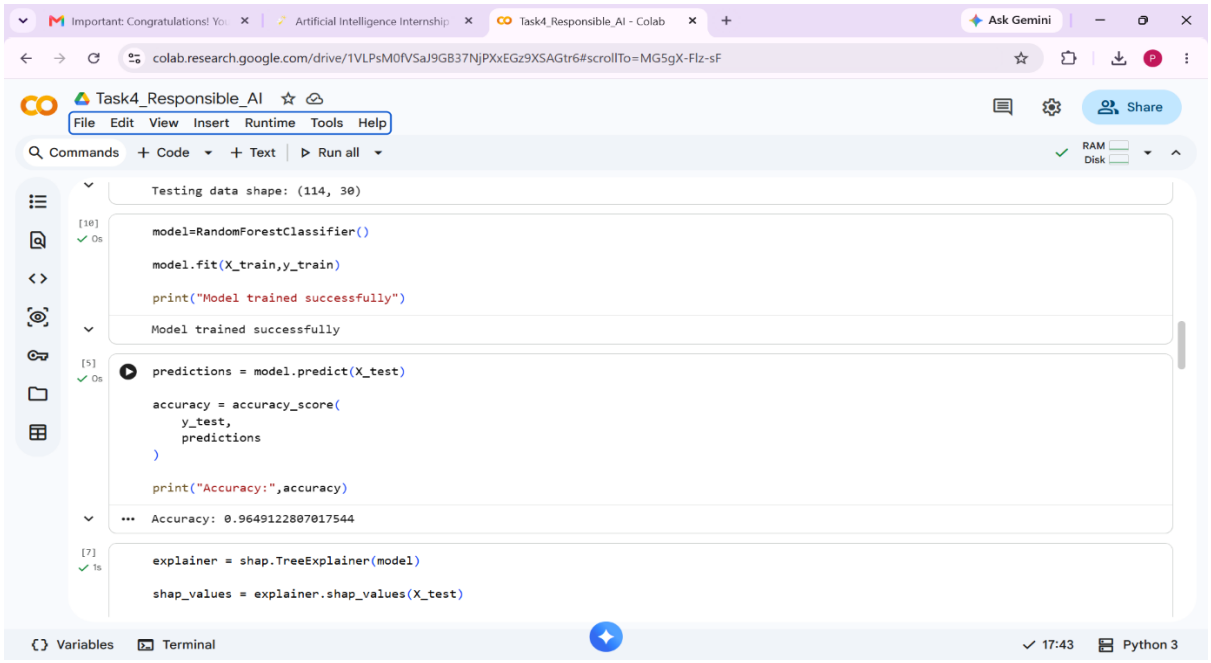
4. Training the Random Forest Model

The Random Forest algorithm was used to train the model. This algorithm builds multiple decision trees and combines their outputs to improve prediction accuracy.



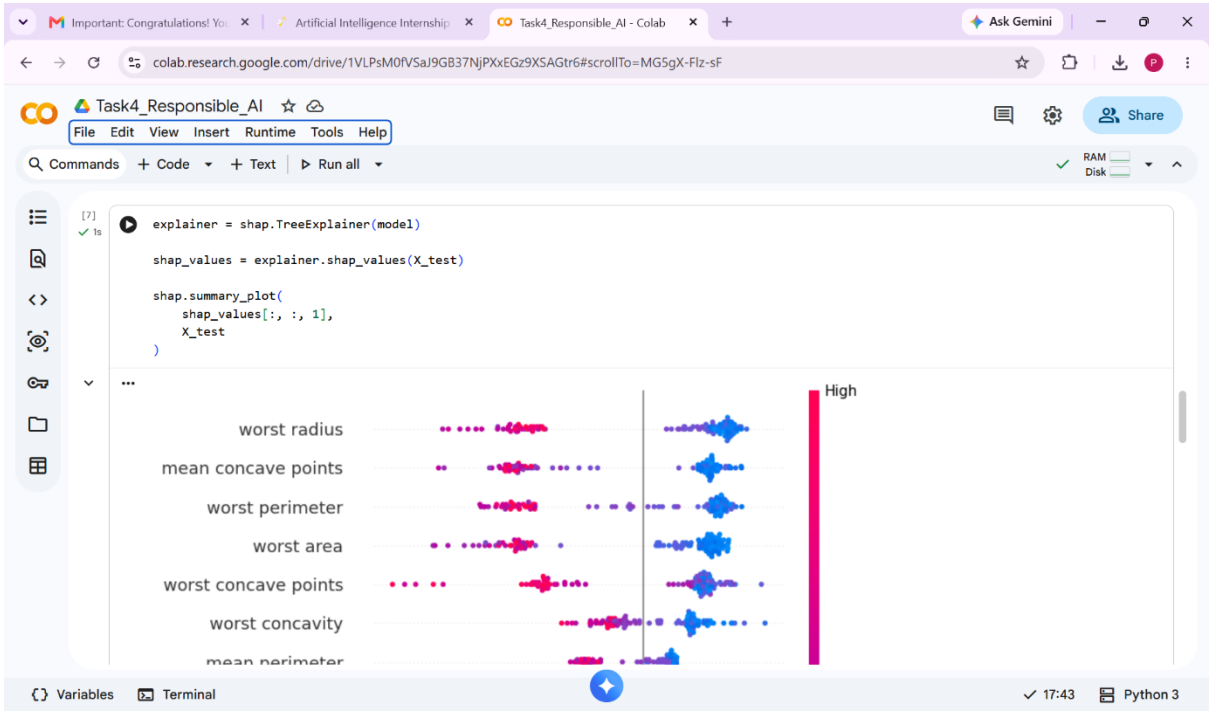
5. Evaluating Model Accuracy

Model performance was evaluated using accuracy score. The obtained result showed that the model was able to classify the dataset with high accuracy.



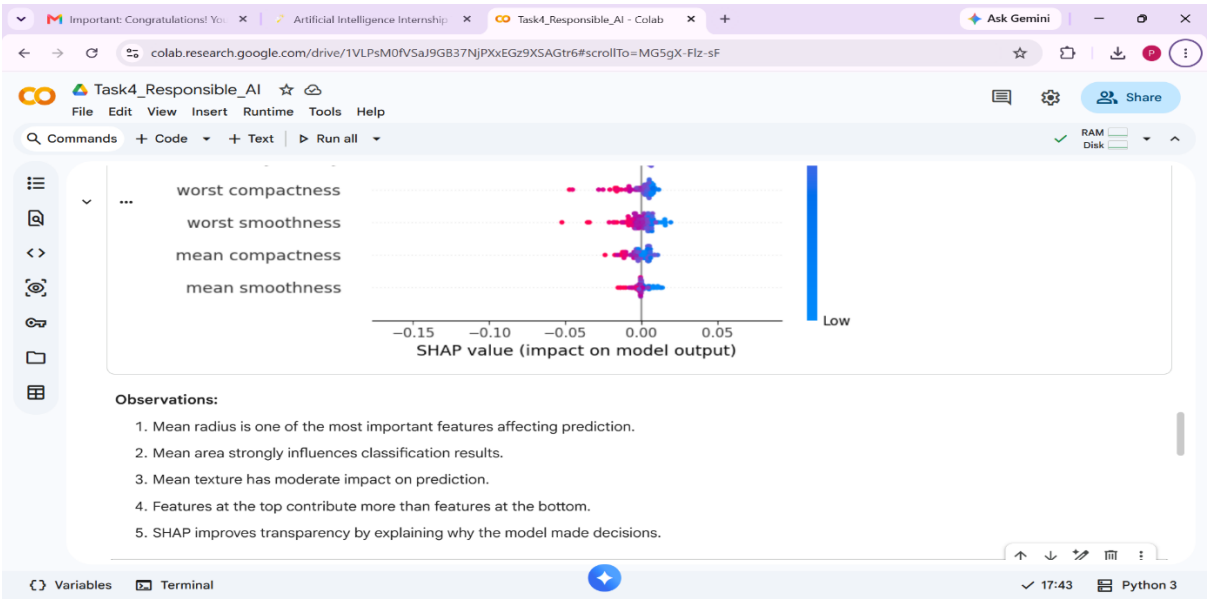
6. Feature Importance Analysis Using SHAP

SHAP analysis was applied to identify the importance of different features affecting predictions. The SHAP summary plot helps visualize the influence of each feature.



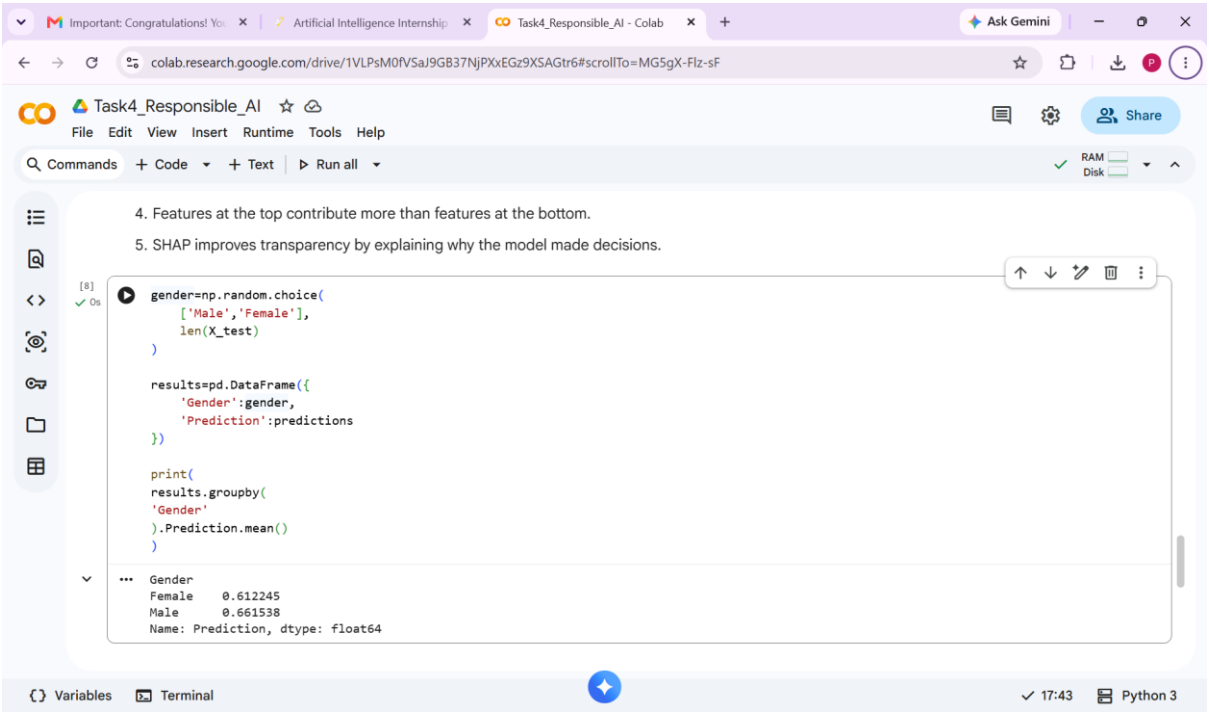
7. Observation Analysis

The generated SHAP graph was analyzed to understand how features contribute to model predictions. Features with larger effects were considered more influential.



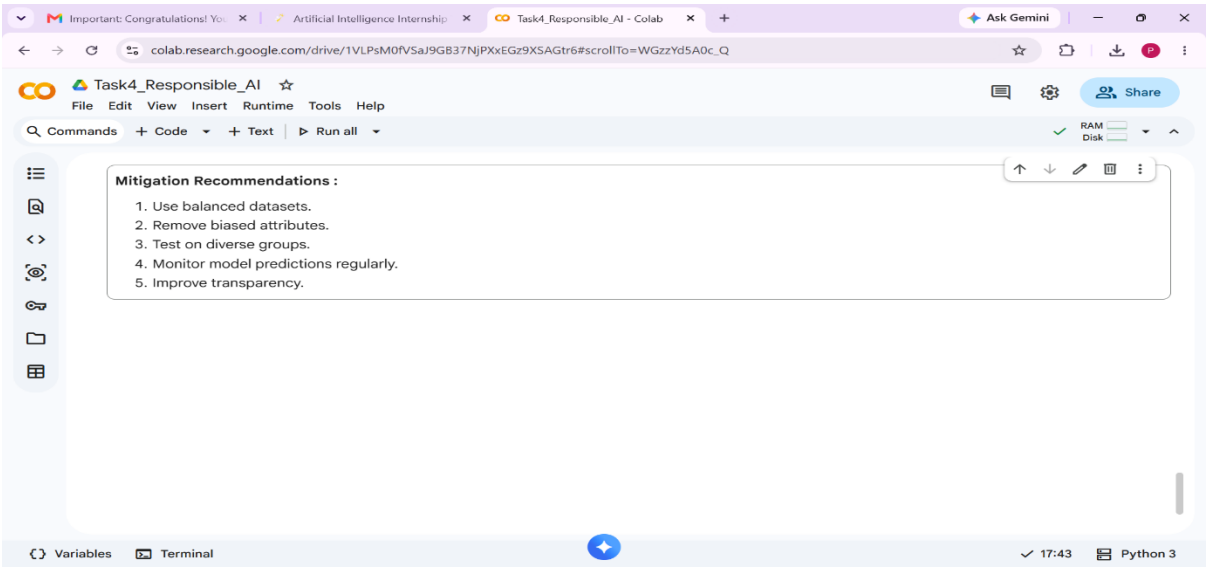
8. Bias Analysis

Bias checking was performed by creating sample groups and comparing prediction outputs. This analysis helps determine whether predictions vary significantly across groups.



9. Mitigation Recommendations

Several mitigation methods were suggested to improve fairness and transparency in the machine learning model.



RESULTS

The Random Forest model achieved approximately 96.49% accuracy. SHAP analysis successfully identified important features influencing predictions. Bias analysis indicated only small differences between groups, showing reasonable fairness.

CONCLUSION

This project successfully demonstrated responsible AI concepts using machine learning and SHAP analysis. The model achieved high prediction accuracy while maintaining transparency through explainable AI techniques. Bias analysis and mitigation recommendations further supported the development of fair and responsible AI systems.

REFERENCES

1. Python Documentation
2. Scikit-learn Documentation
3. SHAP Documentation
4. Google Colab Documentation